> High speed 4-2, 3-2, 2-1 kickdown shifts are prevented
> No EMCC

Warm: Oil temperature at start up above 2.2°C (36°F) and below 27°C (80°F)

> Goes to a Hot schedule above 27°C (80°F) oil temperature
> Normal operation (upshifts, kickdowns, and coastdowns)
> No EMCC

Hot: Oil temperature at start up above 27°C (80°F)

> Goes to a Overheat schedule above 115°C (240°F) oil temperature
> Normal operation (upshifts, kickdowns, and coastdowns)
> Full EMCC, No PEMCC except to engage FEMCC, except at closed throttle at speeds above 113-133 Km/h (70 - 83 MPH)

Overheat: Oil temperature above 115°C (240 °F) or engine coolant temperature above 118°C (244°F)

> Goes to a Hot below 110°C (230°F) oil temperature or a Super Overheat above 115°C (240°F) oil temperature
> Delayed 2-3 upshift 40-51 Km/h (25-32 MPH)
> Delayed 3-4 upshift 66-77 Km/h (41-48 MPH)
> 3rd gear FEMCC from 48-77 Km/h (30-48 MPH)
> 3rd gear PEMCC from 43-50 Km/h (27-31 MPH)

Super Overheat: Oil temperature above 127°C (260°F)

> Goes back to a Overheat below 115°C (240°F) oil temperature
> All a Overheat shift schedules features apply
> 2nd gear PEMCC above 35 Km/h (22 MPH)
> Above 35 Km/h (22 MPH) the torque converter will not unlock unless the throttle is closed (i.e. at 80 Km/h (50 MPH) a 4th FEMCC to 3rd FEMCC shift will be made during a part throttle kickdown or a 4th FEMCC to 2nd PEMCC shift will be made at wide open throttle) or if a wide open throttle 2nd PEMCC to 1 kickdown is made.

Causes for operation in the wrong temperature shift schedule:

Extreme Cold or Cold shift schedule at start up:

> Temperature Sensor circuit.
> Overheat or Super Overheat shift schedule after extended operation:
> Operation in city traffic or stop and go traffic
> Engine idle speed too high
> Aggressive driving in low gear
> Trailer towing in OD gear position (use 3 position (or A/S 3rd) if frequent shifting occurs)
> Cooling system failure causing engine to operate over 110°C (230°F)
> Engine coolant temperature stays low too long - If engine coolant temperature drops below 65°C (150°F), the transmission will disengage EMCC. Extended operation with the EMCC disengaged will cause the transmission to overheat.
> Brake switch issue will cause the EMCC to disengage. Extended operation with the EMCC disengaged will cause the transmission to overheat.
> Transmission fluid overfilled
> Transmission cooler or cooler lines restricted
> Transmission Temperature Sensor circuit

3.2.4 O2 SENSOR (NGC)

The O2 system will with ignition on and engine off have a normalized O2 voltage of around 5 volts as displayed on the DRBIII or measured with a high impedance voltmeter. As the O2 sensor starts generating a signal the voltage will move towards 2.5 volts. The voltage will typically vary between 2.5 volts and 3.5 volts on a normal running engine. The goal voltage is also typically between 2.5 and 3.5 volts. This implies that the 0-volt through 1-volt range that you are used to is still valid, only it is shifted up by a 2.5 volt offset. This 2.5 volt supply is being delivered through the sensor return line.

3.2.5 OTHER CONTROLS

Charging System (NGC)

The charging system is turned on when the engine is started. The Generator Field is controlled by the PCM using a 12-volt high side driver and a body ground. The PCM determines the Generator output voltage by an input from the Battery Temperature Sensor. The PCM applies a longer duty cycle on time to the Generator Field Control circuit when more system voltage is needed. When a lower system voltage is needed, the PCM shortens the duty cycle on time of the high side driver.

Vehicle Speed Control (NGC)

The PCM controls vehicle speed by operation of the speed control servo vacuum and vent solenoids. Energizing the vacuum solenoid applies vacuum to the servo to increase throttle position. Operation of the vent solenoid slowly releases the vacuum allowing throttle position to decrease. A special dump solenoid allows immediate release of throttle position caused by braking, cruise control turn off,